

Course Code: AI401

Course Name: Artificial Intelligence

Semester: Fall 2026

Course Objectives:

Understand the fundamental concepts of Artificial Intelligence, intelligent agents, search techniques, knowledge representation, reasoning, and modern AI applications.

Course Content:

UNIT 1: Introduction to Artificial Intelligence

- History and Applications of AI
- Intelligent Agents
- Types of Agents
- Problem Solving in AI
- AI Environment

UNIT 2: Search Techniques

- State Space Search
- Breadth First Search
- Depth First Search
- Uniform Cost Search
- A* Search Algorithm
- Heuristic Search

UNIT 3: Knowledge Representation

- Propositional Logic
- Predicate Logic
- Semantic Networks
- Frames
- Production Rules

UNIT 4: Machine Learning Basics

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Decision Trees
- Naive Bayes

UNIT 5: Modern AI Applications

- Natural Language Processing
- Computer Vision
- Expert Systems
- Robotics

- Generative AI

Suggested Books:

- Stuart Russell & Peter Norvig – Artificial Intelligence: A Modern Approach
- Elaine Rich & Kevin Knight – Artificial Intelligence

Course Outcomes:

- Understand AI fundamentals and intelligent agents.
- Apply search and reasoning techniques.
- Understand machine learning concepts.
- Analyze real-world AI applications.